

Strategic Quality Management

Science

May, 2026

Strategic Quality Management (A00961)

Short Title:	Strategic Quality Management		
Faculty	Science and Computing		
Department	Science		
Cluster	Quality, Regulatory and Compliance		
Author	MKINSELLA		
Credits	10	Level:	Postgraduate

Description of Module / Aims

This module will provide students with advanced understanding of the theory, practices and implementation issues associated with advanced quality methodologies such as Leadership, Six Sigma and Lean Manufacturing and will also develop competence in a range of technical and management quality tools and how these apply in an industrial environment. Competitive advantage, project management and managing change are key components of applying quality management principles in a industrial setting and will be covered in detail. Validation and calibration of equipment and methods with emphasis on validation documentation will be provided.

Programmes

STAT-0051	Certificate in Advanced Quality Methodologies (WD_SQUAL_MA)	5 / 0 / M
STAT-0051	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 0 / E
STAT-0051	PG Dip in Analytical Science with Quality Management (WD_SANSC_G)	1 / 0 / E

Indicative Content

- Overview of lean practices before and after examples of 5s, Kaizen, Kanban and pull systems
- Reducing cycle times (cycle time versus takt time), identifying bottleneck, process mapping
- Continuous improvement, process variability (statistical approach) and A3 approach to problem solving
- Increasing and utilising capacity and visual management
- Raw material testing, ID testing and streamlining; running samples, cycle time, release of samples, QP, solution, sample and product preparation, stability, out of specification investigation
- Specifications, control of documentation, reference and retention samples, sampling and testing, review of guideline Vs SOP's
- DMAIC (Define, Measure, Analyse, Improve, Control), DFSS, (Design for Six Sigma), TRIZ (theory of Inventive problem-solving)
- Combining lean principles and six sigma practices, review of tools and methodology
- System Thinking: Human Relations at work, participative management, teamwork, risk management systems, quality function deployment
- Measurement of uncertainty, statistical variation, detection and quantification
- Analytical method development, validation of analytical equipment and transfer to QC; life cycle approach, DQ, IQ, OQ and PQ
- Calibration of analytical equipment, optimisation, validation and production, role of standards, calibration management systems and documentation
- Validation documentation, cleaning validation, GAMP, automated and computer system validation, SUPAC, ICH and Pharmacopeia guidelines
- Project management, planning (include cross-function teams, scope communication to non-scientists, implementation and follow-up)
- Managing change in organisation, leadership & motivation and competitive advantage

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate how lean, six sigma and risk management principles can be used to guide lean implementation.
2. Distinguish the key factors for effective motivation of staff applying lean tools and techniques.
3. Justify critical thinking, problem-solving and decision making, project management and the management of change in an industrial environment.
4. Evaluate the effect that routine laboratory operations have on the quality of the final product.
5. Appraise the principles and practices of analytical method and equipment validation and summarise a systematic approach to validation experimentation.
6. Assess the importance and practices of validation activities, documentation and analytical method transfer, including automated and computer systems.

Learning and Teaching Methods

- Lectures.
- Tutorials.
- Presentation.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Presentation</i>	25%	1
<i>In-Class Assessment</i>	25%	5,6
<i>Project</i>	50%	1,2,3,4

Assessment Criteria

<40%: No understanding of the basic concepts of Quality Management. Unable to apply the knowledge given to an industrial context.

40%-59%: Some consideration of the concepts of Quality Management. Some attempt at application of theory to practice.

60%-69%: Show evidence of underpinning knowledge of subject area. Some evidence of evaluation and synthesis of subject area.

70%-100%: Good level of critical analysis, originality of thought and comprehensive knowledge of the subject area. Able to evaluate and synthesise relevant issues.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		36
Tutorial		12
Independent Learning		222

Essential Material

- Bhat, K. Shridhara. *Total Quality Management: text and cases*. Mumbai, India: Himalaya Publishing House, 2010.
- FDA. *Guidance for Industry Process Validation: General Principles and Practices* by U.S. Department of Health and Human Services Food and Drug Administration, .. n/a. 2011.
- FDA. *Guidelines for the development and validation of near infrared (NIR) spectroscopic methods* by An Association of Analytical Chemists within the research based Pharmaceutical Industry, .. n/a. n/a.
- Lal, H. *Organizational Excellence Through Total Quality Management - A Practical Approach* . New Delhi, India: New Age International Publishers, 2008.
- Summers, D.C. *Quality Management; creating and sustaining organizational effectiveness*. 2nd ed. New Jersey, USA: Pearson Prentice Hall, 2008.

Supplementary Material

- "Emerald Database." <http://www.emeraldinsight.com/journals.htm?issn=1351-3036>
- Desai,, Deepali Kishor. *Six Sigma*. Mumbai, India: Global Media, 2010.
- Goldsby, Thomas J. and Robert Martichenko. *Lean Six Sigma Logistics*. Florida, USA: J. Ross Publishing Inc., 2005.
- Henderson, Robin. *Six Sigma Quality Improvement with Minitab*. New Jersey, USA: Wiley, 2011.
- Taghizadegan, Salman. *Essentials of Lean Six Sigma*. Boston: Butterworth-Heinemann, 2006.
- Taylor, Gerald. *Lean Six Sigma Service Excellence H/C: A Guide to Green Belt Certification and bottom line improvement*. Florida, USA: J. Ross Publishing Inc., 2008.